KRISHNA UNNAM

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EDUCATION

University of Illinois Urbana-Champaign - Champaign, IL Graduation: Dec 2027

Bachelor of Science in Bioengineering, Minor in Computer Science & Electrical Engineering

Relevant Coursework: Computational Tools for Biological Data, Linear Algebra for Biomedical Data Science, Biomedical Instrumentation, Introduction to Computing, Signals and Systems in Bioengineering, Preclinical Molecular Imaging

EXPERIENCES

United States Gypsum June 2026 - Aug 2026
Research Intern Libertyville, IL

- Analyzed gypsum composite composition, additive formulations, and processing conditions to optimize reaction rates and performance
- Optimized manufacturing processes targeting a 30% increase in production line output by reducing overall setup time and costs
- Evaluated chemical accelerators to control material behavior and improve overall product performance and process consistency

SakuraMed Feb 2026 - May 2026
Electrical Engineering Intern Champaign, IL

- Reduced PCB footprint by 33% through circuit abstraction and overall design optimization for a cervical screening imaging device
- Implemented AES-128 hardware encryption to secure imaging data while maintaining low-latency and reliable communication
- Integrated electronic components with software and imaging algorithms to improve device performance and simplify data display

Abbott May 2025 - Sep 2025
Technology Intern Champaign, IL

- Developed 3+ cloud data pipelines to automate approval processes, identify inefficiencies, and significantly reduce manual intervention
- Optimized Power BI dashboards to transform large datasets, improving accessibility and key analytical workflow efficiency by 25%
- Digitized a 50+ question qualification form, streamlining data processing and reducing submission errors across operational workflows

PROJECTS

Explainable AI for Automated ECG Diagnosis and Interpretation Nov 2025 - Present
Machine Learning Engineer Champaign, IL

- Developed a CNN-LSTM pipeline for ECG arrhythmia detection and classification, achieving 85% binary classification accuracy
- Engineered a preprocessing pipeline for PQRST-aligned ECG segmentation and feature extraction to improve model stability
- Designed a SHAP-based explainability framework to identify waveform regions driving predictions and improve model interpretability

CARDIOgraph Sep 2025 - Present
Biomedical Data Engineer Champaign, IL

- Designed a data ingestion pipeline integrating 5+ biomedical databases to build a Neo4j graph of 1,000+ drug–disease relationships
- Processed 9,000+ graph nodes to generate embeddings using Node2Vec and DVGAE for cardiotoxic drug–disease link prediction
- Developed an interface for querying biomedical datasets across 4 cardiovascular conditions for cardiotoxicity pathway analysis

Heart Attack Detection Sept 2024 - May 2025
Electronics Engineering Lead Champaign, IL

- Developed a real-time ECG monitoring system that detected cardiac irregularities with 90% accuracy using ST-segment analysis
- Optimized 6-lead electrode placement across intercostal gaps to improve signal-to-noise ratio and enhance arrhythmia detection
- Applied signal processing and circuit abstraction techniques to amplify, denoise, and analyze ECG signals for real-time monitoring

LEADERSHIP & ACTIVITIES

Illinois Medical Advancements through Design and Engineering (i-MADE) Sep 2024 - Present
Vice President of Finance Champaign, IL

- Manage recruitment by evaluating 100+ qualified applicants and coordinating interviews to expand organizational membership
- Oversee organizational budget and resource allocation by tracking expenses, allocating funds, and maintaining financial records
- Coordinate engineering projects and present solutions at UIUC's Engineering Open House and Undergraduate Research Symposium

FlexAble Feb 2026 - Present
Co-Founder/Engineering Lead Champaign, IL

- Engineered an Arduino-controlled glove integrating servo actuation and embedded electronics for joint stabilization and grip assistance
- Programmed an Arduino control system to process real-time sensor feedback and dynamically adjust actuator-assisted grip force
- Pitched the prototype at the COZAD New Venture Challenge, winning the $10,000 Award for innovation and clinical impact

SKILLS & INTERESTS

Manufacturing: Process Optimization | Manufacturing Systems | Quality Engineering | Fusion 360 | KiCAD | Simulink | Arduino | PCB
Data Analytics: Python (NumPy, Pandas) | SQL | MATLAB | Power BI | Excel | ETL | Time-Series Analysis | Data Visualization
Programming & Imaging: Java | C++ | JavaScript | HTML/CSS | React.js | Node.js | OpenCV | TensorFlow | ImageJ | FFT | PCA
Interests: Process Improvement | Manufacturing Systems | Quality Engineering | Operations Management | Automation | Production